Coffee Price modeling in R

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1. Objective

This report prices a six-month coffee futures contract and a European call option on coffee. It combines three quantitative methods:

- The cost of carry model, to price the futures contract;
- The Black-Scholes model, to price the call option analytically;
- Monte Carlo simulation, to price the same option numerically and validate the analytical result.

2. Market parameters

```
S_0 <- 1.2    #Coffee spot price ($/lb)
r <- 0.02      #Annual risk-free rate (2%)
d <- 0.01     #Annual storage cost (1%)
t <- 0.5      #Time to maturity T (years)
X <- 1.25     #Strike price ($/lb)
sigma <- 0.25    #Annual volatility (25%)

market_data <- data.frame(
  Parameter = c("Spot price (S_0)", "Risk-free rate (r)", "Storage cost (d)",
                "Maturity (T)", "Strike price (X)", "Volatility (sigma)"),
  Value = c(S_0, r, d, t, X, sigma)
)

knitr::kable(market_data,
             caption = "Market Data and Assumptions")
```

Table 1: Market Data and Assumptions

Parameter	Value
Spot price (S_0)	1.20
Risk-free rate (r)	0.02
Storage cost (d)	0.01
Maturity (T)	0.50
Strike price (X)	1.25
Volatility (sigma)	0.25

The option is out of the money at inception, since the spot price ($1.20) is below the strike price ($1.25)

3. Pricing functions

The pricing models are defined once as functions, so the same code is reused in the base case and in the sensitivity analysis.

3.1 Step 1: Cost of carry model

```
b <- r + d                #risk-free interest rate + storage cost
F_t  <- S_0 * exp(b*t)     #Future price

sprintf("Future price : $%.4f per pound", F_t)
```

```
## [1] "Future price : $1.2181 per pound"
```

3.2 Step 2: Black-Scholes model of a European call on a non-dividend-paying asset

```
d1 <- (log(S_0/X) + (r+(sigma^2)/2)*t)/(sigma*sqrt(t)) #BS d1 parameter
d2 <- d1 - (sigma*sqrt(t))                              #BS d2 parameter

Call_bs <- (S_0 * pnorm(d1)) - (X*exp(-r*t)*pnorm(d2))   #Call option price


sprintf("Call price from BS : $%.4f per pound", Call_bs)
```

```
## [1] "Call price from BS : $0.0684 per pound"
```

3.3 Step 3: Monte Carlo simulations of the European call under risk-neutral GBM

```
n_sim <- 10000      #Number of simulations
set.seed(42)

Z <- rnorm(n_sim, mean = 0, sd = 1)  #Standard Normal draw

S_T <- S_0 * exp(((r - (sigma^2)/2)*t) + (sigma*sqrt(t)*Z))  #Geometric Brownian Motion
payoff <- pmax(S_T - X,0)     #Payoff for each simulation

Call_mc <- exp(-r*t) * mean(payoff)    #Call price from Monte Carlo Simulation

se <- (sd(exp(-r*t) * payoff))/sqrt(n_sim)   #Standard error (CLT)

sprintf("Call price from MC : $%.4f per pound (standard error:$%.4f)", Call_mc,se)
```

```
## [1] "Call price from MC : $0.0677 per pound (standard error:$0.0013)"
```

Comparison of the two pricing methods: Black-Scholes and Monte Carlo simulations

```
pricing_comparison <- data.frame(
  Method = c("Black-Scholes", "Monte Carlo"),
  Call_Price = c(Call_bs, Call_mc)
)

knitr::kable(pricing_comparison,
             digits = 4,
             col.names = c("Method", "Call price (USD/lb)"),
             caption = "Call option pricing")
```

Table 2: Call option pricing

Method	Call price (USD/lb)
Black-Scholes	0.0684
Monte Carlo	0.0677

```
pricing_error <- abs(Call_mc - Call_bs)      #Absolute pricing difference

sprintf("Absolute pricing difference: $%.4f per pound", pricing_error)
```

```
## [1] "Absolute pricing difference: $0.0007 per pound"
```

3.4 Step 4: Sensitivity analysis

3.4.1 Sensitivity to the coffee spot price

```
spot_scenarios <- seq(1.00, 1.40, by = 0.05)

# Black-Scholes pricing function
bs_call <- function(S_0, K, r, sigma, t) {
  d1 <- (log(S_0/K) + (r + sigma^2/2)*t) /
        (sigma*sqrt(t))
  d2 <- d1 - sigma*sqrt(t)

  S_0*pnorm(d1) - K*exp(-r*t)*pnorm(d2)
}

# Calculate call prices for different spot prices
call_prices <- bs_call(
  spot_scenarios, X, r, sigma, t
)

sensitivity_table <- data.frame(
  Spot_Price = spot_scenarios,
  Call_Price = call_prices
)
```

```r
knitr::kable(
  sensitivity_table,
  digits = 4,
  col.names = c("Spot price (USD/lb)", "Call price (USD/lb)"),
  caption = "Call Option Sensitivity to Spot Prices"
)
```

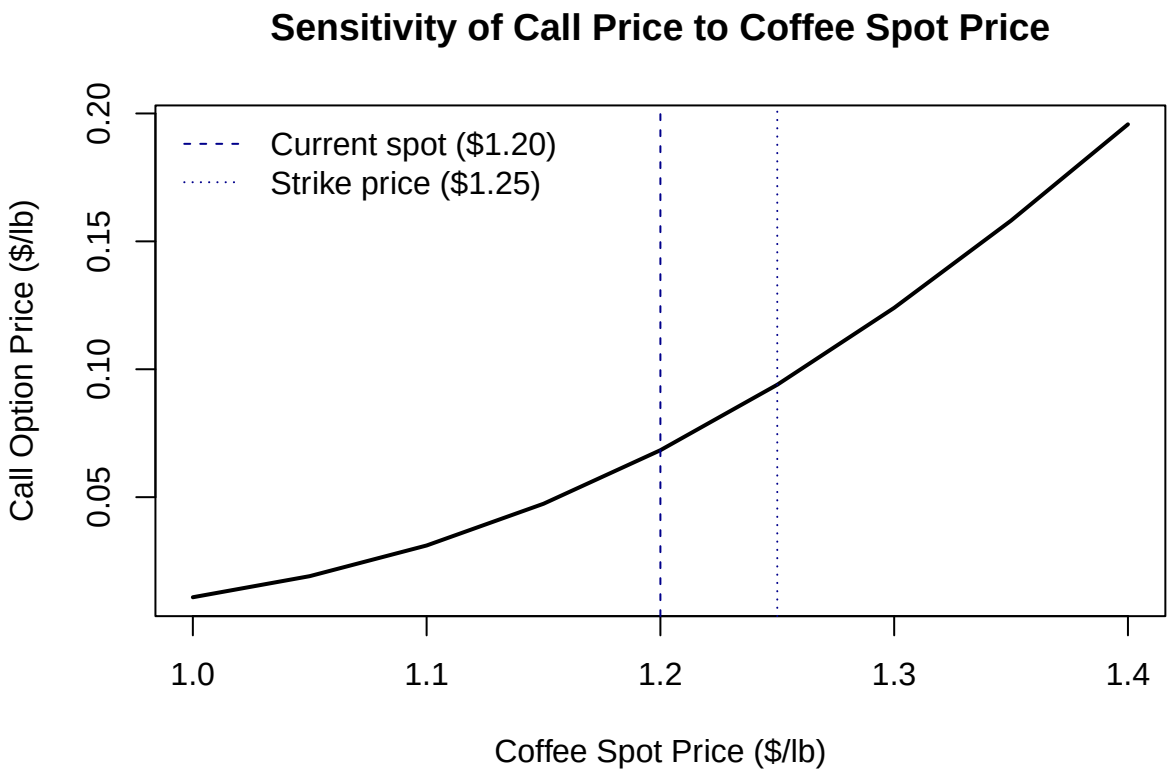
Table 3: Call Option Sensitivity to Spot Prices

Spot price (USD/lb)	Call price (USD/lb)
1.00	0.0109
1.05	0.0191
1.10	0.0311
1.15	0.0474
1.20	0.0684
1.25	0.0940
1.30	0.1240
1.35	0.1581
1.40	0.1957

```r
# Visualisation

plot(spot_scenarios,call_prices, type = "l", lwd = 2,
  xlab = "Coffee Spot Price ($/lb)",
  ylab = "Call Option Price ($/lb)",
  main = "Sensitivity of Call Price to Coffee Spot Price"
)

abline(v = S_0, lty = 2, col ="blue4")
abline(v = X, lty = 3, col = "blue4")
legend("topleft", legend = c("Current spot ($1.20)", "Strike price ($1.25)"),
                            lty = c(2,3), col = "blue4", bty = "n")
```



Interpretation : As the underlying coffee price increases, the call option becomes more valuable (call price rises with the spot price from \$0.0109 at \$1.00 to \$0.1957 at \$1.40 per pound). The sensitivity analysis reveals a positive and convex relationship between coffee spot prices and call option premiums. The convex shape of the curve reflects the option's positive gamma, as its delta increases when the option moves from out of the money to in the money.

3.4.2 Sensitivity to volatility

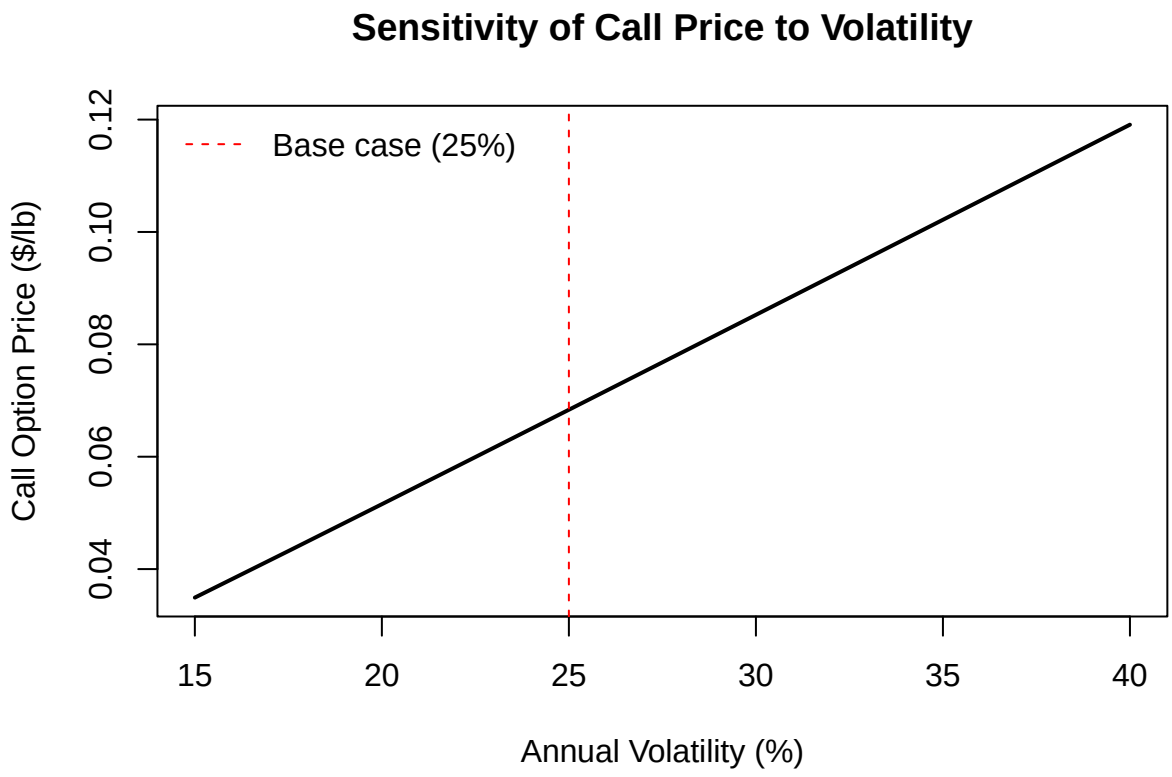
```
vol_scenarios <- seq(0.15, 0.40, by = 0.025)

vol_call_prices <- bs_call(
  S_0, X, r, vol_scenarios, t
)


#Visualisation

plot(vol_scenarios * 100, vol_call_prices, type = "l", lwd = 2,
  xlab = "Annual Volatility (%)",
  ylab = "Call Option Price ($/lb)",
  main = "Sensitivity of Call Price to Volatility"
)
```

```r
abline(v = sigma * 100, lty = 2, col = "red")
legend("topleft", legend = "Base case (25%)", lty = 2, col = "red", bty = "n")
```



Interpretation : The call price rises with volatility from \$0.0349 at 15% to \$0.1191 at 40%. The positive relationship reflects the option's positive vega. The higher volatility widens the range of possible coffee prices at maturity, which increases the chance that the option ends in the money while the loss remains limited to the premium. Over this range, the relationship is close to linear

4. Market factors and pricing implications

```r
market_factors <- data.frame(
  m_factor = c("Lower supply (poor harvest in Brazil or Vietnam)",
               "Stronger global demand",
               "weather risk (frost, drought, excess rain)",
               "Geopolitical risk (trade, restrictions, shipping disruptions)"
               ),
  channel = c("Higher spot price",
              "Higher spot price",
              "Higher volatility",
              "Higher storage costs and volatility"
              ),
  Effect_pricing = c("Higher futures and call prices",
```

```
                      "Higher futures and call prices",
                      "Higher call price (positive vega)",
                      "Higher futures price; Higher call price"
                      )
)

knitr::kable(market_factors,
             col.names = c("Market factors", "Main channel", "Effect on pricing"),
             caption = "Commodity Market Risk Analysis")
```

Table 4: Commodity Market Risk Analysis

Market factors	Main channel	Effect on pricing
Lower supply (poor harvest in Brazil or Vietnam)	Higher spot price	Higher futures and call prices
Stronger global demand	Higher spot price	Higher futures and call prices
weather risk (frost, drought, excess rain)	Higher volatility	Higher call price (positive vega)
Geopolitical risk (trade, restrictions, shipping disruptions)	Higher storage costs and volatility	Higher futures price; Higher call price

These factors can interact, potentially amplifying price movements and market uncertainty.

5. Conclusion

The cost of carry model estimates a six-month coffee futures price of approximately $1.2181 per pound. The Black-Scholes model values the European call at $0.0684 per pound, and the Monte Carlo simulation gives $0.0677 per pound. The close agreement between the analytical and Monte Carlo estimates provides a numerical consistency check for the pricing implementation.

The sensitivity analysis shows that the option premium increases with both the underlying spot price and volatility. Supply and demand shocks mainly affect the premium through the spot price, while weather and geopolitical risk mainly affect it through volatility.

Overall, the analysis highlights the importance of combining quantitative valuation techniques with an analysis of commodity market fundamentals for a more complete view when assessing pricing and risk.